# CliniCause: Constructing Clinically Grounded Observational Testbeds for Causal Analysis

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## Abstract

Causal estimators are commonly evaluated using synthetic, semi-synthetic, or experimentally anchored benchmarks whose ground truth depends on researcher-specified data-generating or sampling mechanisms. We introduce *CliniCause*, a complementary framework for constructing observational causal-analysis testbeds without generating patient records, exposure assignments, or outcomes. CliniCause uses literature-grounded LLM elicitation to define clinically motivated proxy exposures and a causal graph, then instantiates this representation on source-observed longitudinal ICU records from MIMIC-III and PhysioNet 2012. We evaluate the resulting resources through proxy recoverability, mortality prediction, cross-estimator agreement, permutation tests, omitted-variable sensitivity analysis, and comparison with clinical literature. The results show broad predictive and cross-estimator consistency alongside meaningful sensitivity to modest unobserved confounding, demonstrating the value of real observational testbeds for exposing estimator fragility that controlled benchmarks may not reveal.

## 1 Introduction

Causal inference asks how outcomes would change under alternative actions, rather than only which outcomes can be predicted. This distinction is central to medicine and public policy: an accurate forecast may identify who is at risk, but it does not establish which action is needed. Modern causal machine-learning methods support flexible adjustment and heterogeneous-effect estimation (Chernozhukov et al. 2018; Wager and Athey 2018; Shalit, Johansson, and Sontag 2017; Louizos et al. 2017), but evaluating them is unusually difficult. For each individual, the counterfactual outcome needed to score an estimated effect is unobserved.

Because observational data do not reveal counterfactual outcomes, they do not provide a direct answer key for evaluating causal estimators. They also leave the true treatment-assignment process and the extent of unmeasured confounding unknown. The safest experimental sandbox is therefore fully synthetic data: the evaluator specifies the covariates, treatment assignment, potential outcomes, confounding, overlap, and effect heterogeneity. Assumptions such as ignorability or positivity can be satisfied, violated, or gradually stressed under complete control, while the true causal effect remains known.

That control creates an important limitation. The evaluator determines not only the answer but also the difficulty of recovering it. A method may perform well because its assumptions align with the selected assignment model, response surface, confounding structure, or overlap regime. Semi-synthetic benchmarks move closer to observed data by retaining empirical covariates while simulating treatment assignments, potential outcomes, or both. RCT-derived and empirically anchored designs preserve still more of the observed system, but continue to obtain an answer key through researcher-controlled sampling, comparison, or generation mechanisms. Across this spectrum, greater control makes accuracy measurable, while greater observational fidelity makes the causal target less knowable.

Several widely used causal-ML resources illustrate this spectrum of constructed evaluation. IHDP retains empirical covariates while simulating outcomes, whereas ACIC uses empirical covariates within diverse simulated treatment and outcome mechanisms; Dragonnet, for example, evaluates 101 retained datasets drawn from 63 ACIC 2018 data-generating settings (Shalit, Johansson, and Sontag 2017; Shi, Blei, and Veitch 2019). Influence-function validation similarly compares models across 77 semi-synthetic competition datasets sharing one empirical feature source but using distinct simulated assignments and outcomes (Alaa and van der Schaar 2019). More empirically grounded designs retain additional observed components: observational sampling from randomized trials preserves real trial variables but induces confounding through researcher-specified selection (Gentzel, Pruthi, and Jensen 2021); Jobs/LaLonde combines experimental treated subjects with external non-experimental controls, making comparability depend on population-selection and harmonization choices; and Credence learns from an empirical distribution, but generates data under user-specified effects and confounding (Parikh et al. 2022). These designs are increasingly sophisticated and provide valuable causal answer keys. Nevertheless, measurable accuracy still depends on an experimental anchor or a researcher-constructed causal, sampling, or comparison mechanism. They therefore cannot by themselves show how estimators behave when exposure prevalence, missingness, support, and measurement

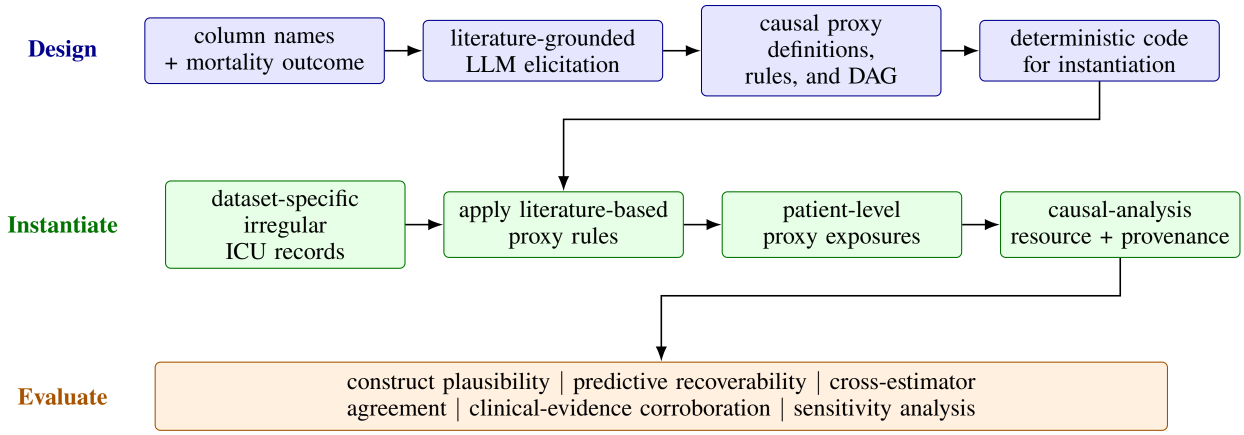


Figure 1: The three stages of CliniCause. Schema-level clinical knowledge is used to define causal proxy states, literature-based rules, and a DAG. Deterministic code instantiates this representation on dataset-specific ICU records, after which the resulting causal-analysis resources are evaluated. Critically, the LLM receives schema information rather than patient-level records.

processes arise from routine practice rather than from the evaluation design.

## Our Contribution.

We propose shifting researcher intervention in causal evaluation from the data-generating process to the *representation layer*. Consider estimating the effect of ICU shock on mortality. A synthetic benchmark specifies how shock is assigned and how mortality is generated, thereby providing a known answer under researcher-selected mechanisms. Our alternative preserves the source-observed measurements, clinical practice, missingness, empirical support, and mortality, while explicitly defining how those records become exposures, covariates, cohorts, and adjustment assumptions.

We instantiate this approach through *CliniCause*, an evidence-tracked pipeline applied to MIMIC-III and PhysioNet 2012 (Johnson et al. 2016; Silva et al. 2012). ChatGPT was given the available column names and mortality outcome, but not patient records, distributions, or estimated effects. It was instructed to identify literature-supported clinical states treated as causal determinants of mortality, operationalize them using measurement cutoffs and temporal rules, and propose a DAG relating the states, observed variables, measurement processes, and mortality. Each proposed proxy, cutoff, and edge was required to include a clinical rationale and supporting source, making the representation inspectable and replaceable. The LLM thus served as a knowledge-elicitation assistant, not as a source of causal ground truth or a mechanism for generating ground truth from the observational data.

Applied to 26,845 MIMIC-III and 7,993 PhysioNet records, CliniCause produced resources with nine and ten proxy exposures, respectively, including renal and hepatic dysfunction, shock, respiratory failure, cardiac strain, and sepsis-related burden.

Although the represented clinical states were selected because the medical literature treats them as causal determinants of mortality, this construct-level evidence does not establish causal identification in these particular datasets. The operational rules, cutoffs, temporal aggregation, cohort definitions, and DAG-based adjustment sets may still introduce measurement error, selection bias, model misspecification, or unmeasured confounding. Because the resulting resources therefore have no known causal answer, we evaluate their construction through converging evidence. Informal ICU-physician consultation assesses construct plausibility; temporal models (STraTS, GRU, GRU-D, and TCN) tested proxy recoverability from the irregular time series using three-fold cross-validation; three causal estimators assess effect-pattern stability; and published clinical evidence provides external directional and scale comparisons. Across 19 prespecified dataset–exposure comparisons, CausalForestDML and LinearDML agreed in direction in all 19, while CausalPFN agreed in 18. Comparison with the clinical literature corroborated most estimated patterns while also identifying several discrepancies for further investigation.

Our contributions, illustrated in Figure 1, can be summarized as follows:

1. An evidence-tracked pipeline that converts irregular clinical records into explicit causal-analysis representations while preserving observed patient, support, and outcome processes;
2. Two resources derived from MIMIC-III and PhysioNet 2012, containing 19 clinically motivated proxy-exposure analyses with inspectable cohort, graph, adjustment, and provenance assumptions; and
3. An evaluation framework based on construct plausibility, proxy recoverability, cross-estimator agreement, permutation checks, omitted-variable sensitivity analysis, and clinical-literature corroboration. Its application shows that otherwise concordant double machine learning (DML) estimates can remain fragile to modest unobserved confounding, demonstrating the value of the proposed resources for exposing weaknesses that agreement alone would conceal.

## 2 Related Work

**Clinical representation.** Electronic phenotyping converts heterogeneous health records into computable states using expert rules and learned models (Banda et al. 2018; Ratner et al. 2020). GRU-D models informative missingness, while STraTS represents irregular observations as time–variable–value events (Che et al. 2018; Tipirneni and Reddy 2022). CliniCause differs by first eliciting clinically and causally motivated proxy states and operational rules, and then using temporal models to test their recoverability.

**LLMs and causal knowledge.** Medical LLMs encode substantial clinical knowledge but require verification (Singhal et al. 2023). Prior work uses LLMs, clinical literature, or observational associations to construct causal graphs, while evaluations identify sensitivity to memorization and prompting (Roy et al. 2025; Feng et al. 2025; Nordon et al. 2019). CliniCause withholds patient records from the LLM and instead uses schema-level information to elicit literature-grounded proxy definitions, cutoffs, temporal rules, and their DAG.

**Causal estimation and evaluation.** Double machine learning and causal forests support flexible nuisance estimation and heterogeneous effects (Chernozhukov et al. 2018; Wager and Athey 2018; Athey, Tibshirani, and Wager 2019), while CausalPFN amortizes estimation across simulated data-generating processes (Meresht et al. 2025). Existing evaluations use semi-synthetic, RCT-derived, or empirically anchored benchmarks, including IHDP, ACIC, Jobs/LaLonde, OSRCT, and Credence (Alaa and van der Schaar 2019; Gentzel, Pruthi, and Jensen 2021; Parikh et al. 2022). These provide experimental or constructed answer keys; CliniCause instead evaluates estimators under a knowledge-grounded representation of source-observed clinical data.

**Observational clinical causality and positioning.** Causal EHR studies require explicit cohorts, exposures, outcomes, time zero, and adjustment assumptions (Hernan and Robins 2016; Hernan and Taubman 2008; Bica et al. 2021). Such representations are usually reconstructed for one clinical question. CliniCause packages multiple clinically motivated exposures, proxy rules, DAGs, adjustment sets, and provenance within a shared resource. To our knowledge, prior work has not combined schema-only LLM knowledge elicitation, deterministic instantiation on longitudinal records, and evaluation through proxy recoverability, estimator agreement, and clinical-literature corroboration.

## 3 Methodology

In this section, we describe the CliniCause construction pipeline summarized in Figure 1: resource definition, knowledge-driven proxy and graph design, patient-level instantiation, and provenance tracking. The subsequent clinical-evidence validation is described in detail in Section 5.3.

### 3.1 Clinical Setting and Resource Contract

Let $d \in \{\mathrm{MIMIC}, \mathrm{PhysioNet}\}$ index the source database and let

$$\mathcal{I}_d = \{1, \dots, N_d\}$$

denote its set of analysis records. An analysis record is an ICU stay or source time-series record, not necessarily a unique patient. For record $i \in \mathcal{I}_d$, the irregular clinical observations are

$$\mathcal{X}_i^{(d)} = \{(t_{ij}, v_{ij}, x_{ij})\}_{j=1}^{n_i},$$

where $t_{ij}, v_{ij}$, and $x_{ij}$ denote the observation time, measured variable, and recorded value. A linked record-level view supplies the canonical identifier $\mathrm{id}_i^{(d)}$, baseline covariates $\mathbf{C}_i^{(d)}$, and source-observed in-hospital mortality outcome $Y_i^{(d)}$. We write the complete source input as

$$\mathcal{D}_d^{\mathrm{src}} = \left\{ \left( \mathrm{id}_i^{(d)},\, \mathcal{X}_i^{(d)},\, \mathbf{C}_i^{(d)},\, Y_i^{(d)} \right) :\ i \in \mathcal{I}_d \right\}.$$

MIMIC-III and PhysioNet 2012 are processed independently (Johnson et al. 2016; Silva et al. 2012); their records, proxy semantics, graphs, and estimates are never pooled.

CliniCause is a factored transformation with separate knowledge-design and patient-instantiation stages. Let $\mathcal{S}_d$ denote the source schema available to the design stage and let $\mathcal{L}$ denote external clinical knowledge and literature. The knowledge function

$$\mathcal{K}_d : \left( \mathcal{S}_d,\, Y^{(d)},\, \mathcal{L} \right) \longmapsto (\mathcal{Q}_d,\, G_d)$$

constructs a set of operational proxy definitions $\mathcal{Q}_d$ and a dataset-specific DAG $G_d$. The LLM participates only in $\mathcal{K}_d$ and has no access to $\mathcal{D}_d^{\mathrm{src}}$.

A deterministic instantiation function $\mathcal{F}_d$ then applies the accepted proxy definitions and DAG to $\mathcal{D}_d^{\mathrm{src}}$. This separation ensures that schema-level knowledge elicitation occurs without patient-level access, while patient records enter only during deterministic instantiation. The resulting causal-analysis resource is

$$\mathcal{R}_d = \left( \mathcal{I}_d,\, \widetilde{\mathbf{Z}}^{(d)},\, \mathbf{Y}^{(d)},\, \mathbf{C}^{(d)},\, G_d,\, \{\mathcal{A}_k^{(d)}\}_{k=1}^{K_d},\, \mathrm{Prov}_d \right),$$

where $\widetilde{\mathbf{Z}}^{(d)}$ contains the instantiated proxy exposures, $\mathbf{Y}^{(d)}$ contains the source-observed outcomes, $\mathbf{C}^{(d)}$ contains the baseline covariates, $\mathcal{A}_k^{(d)}$ is the recorded adjustment set for exposure $k$, and $\mathrm{Prov}_d$ contains configuration, artifact, and lineage metadata. Thus, the irregular time series $\{\mathcal{X}_i^{(d)}\}_{i \in \mathcal{I}_d}$ are source inputs, whereas $\widetilde{\mathbf{Z}}^{(d)}$ is the constructed tabular exposure interface.

The shared resource contract standardizes analytical roles rather than imposing a common clinical vocabulary. The admitted resources contain 26,845 MIMIC-III records with $K_{\mathrm{MIMIC}} = 9$ proxy exposures and 7,993 PhysioNet records with $K_{\mathrm{PhysioNet}} = 10$.

### 3.2 Knowledge-Driven Proxy and Graph Design

The knowledge-driven design stage first defines proxy variables and their operational rules, and only then organizes those constructs into a causal DAG. For source $d$, the LLM

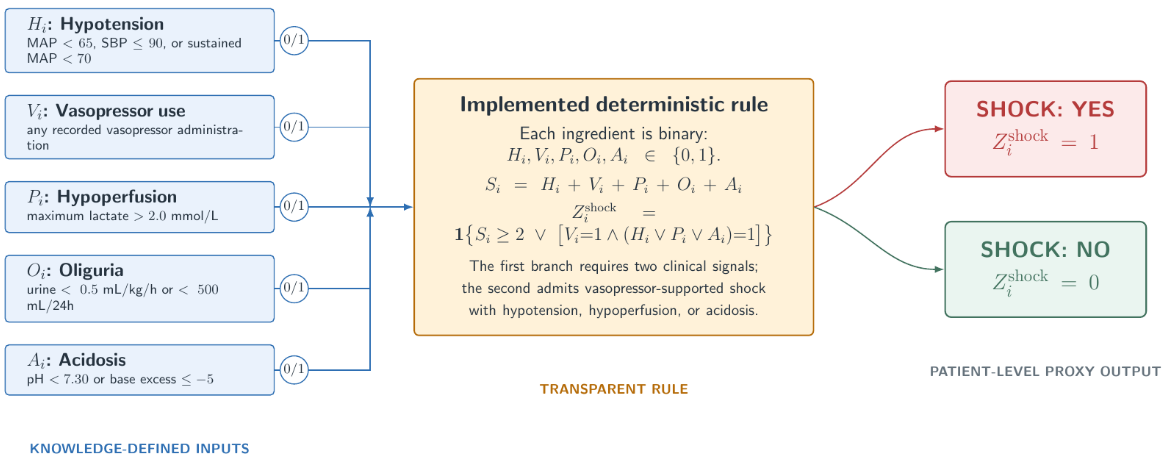


Figure 2: Example of ChatGPT-assisted, knowledge-driven proxy construction for the shock exposure. Five binary clinical indicators $(H_i, V_i, P_i, O_i, A_i)$ are derived from the corresponding patient time series and combined by an explicit deterministic rule to produce the patient-level proxy $Z_i^{\text{shock}}$. ChatGPT proposes candidate indicators, cutoffs, and combination logic at design time; the authors inspect and encode the accepted specification.

receives the available column names $\mathcal{S}_d$ and target outcome $Y$, but no patient rows, empirical distributions, exposure prevalences, or effect estimates. The elicitation step was conducted interactively through the ChatGPT desktop interface using the prompt at the end of the Supplementary file. Figure 2 illustrates how five raw time-series measurements are mapped to the shock proxy. More generally, the knowledge-elicitation stage is represented as

$$\mathcal{K}_d : (\mathcal{S}_d, Y^{(d)}) \longmapsto \left\{ \left( Z_k^{(d)}, r_k^{(d)}, \mathcal{L}_k^{(d)} \right) \right\}_{k=1}^{K_d},$$

where $Z_k^{(d)}$ is a candidate clinical state, $r_k^{(d)}$ contains its cutoffs and temporal rules, and $\mathcal{L}_k^{(d)}$ records its clinical rationale and supporting literature. ChatGPT was instructed to ground every proposed proxy and cutoff in published clinical evidence. The authors inspected the proposed rationales and translated the selected definitions into deterministic source code. Because ChatGPT received the schema and target outcome but neither patient records nor empirical effect estimates, this stage could not select proxy definitions according to their observed MIMIC-III or PhysioNet effects. After fixing the proxy set, the LLM proposes a directed graph

$$G_d = (V_d, E_d), \qquad V_d = \mathbf{C}^{(d)} \cup \mathbf{Z}^{(d)} \cup \{Y^{(d)}\},$$

with a clinical rationale and supporting source for each proposed edge $(u \to v) \in E_d$.

### 3.3 Patient-Level Instantiation and Resource Assembly

Deterministic code applies the accepted, ChatGPT-elicited proxy definitions to each ICU record. The irregular measurements are first summarized over their available observations, after which the literature-grounded cutoffs and temporal rules are evaluated. Figure 2 illustrates this process for shock: observed evidence from blood pressure, lactate, vasopressor use, urine output, and acid–base measurements is combined according to a prespecified rule to produce a binary shock proxy. The same procedure is applied independently to every proxy and dataset.

Missing measurements are not numerically imputed before rule evaluation. A threshold condition contributes evidence only when its required measurement is observed; otherwise, that condition is false. Similarly, treatment or care-process indicators contribute evidence only when recorded as positive. Consequently, a proxy is assigned 1 only when the available record contains sufficient abnormal evidence to satisfy its rule. A value of 0 therefore means that the proxy rule was not satisfied by the observed record, not that the underlying clinical state was definitively absent. This distinction is important because ICU measurement availability reflects both patient state and clinical practice.

For record $i$ in dataset $d$, rule-based instantiation produces

$$Z_{ik}^{(d)} = r_k^{(d)}\left(\mathcal{X}_i^{(d)}\right), \qquad k = 1, \ldots, K_d,$$

where $r_k^{(d)}$ is the accepted deterministic rule and $\mathcal{X}_i^{(d)}$ is the irregular source time series. The resulting estimator-agnostic resource row is

$$R_i^{(d)} = \left( \mathrm{id}_i^{(d)}, \mathbf{Z}_i^{(d)}, Y_i^{(d)}, \mathbf{C}_i^{(d)} \right),$$

where $\mathbf{Z}_i^{(d)}$ contains the rule-derived proxy exposures, $Y_i^{(d)}$ is source-observed in-hospital mortality, and $\mathbf{C}_i^{(d)}$ contains the observed covariates. For analysis of proxy $k$, $Z_{ik}^{(d)}$ becomes the exposure and the dataset-specific DAG supplies its recorded adjustment set.

## 4 Empirical Evaluation

Without known causal ground truth, CliniCause requires a converging-evidence evaluation scheme. This section de-

| Resource | Records | Proxies | Outcome |
|---|---|---|---|
| MIMIC-III | 26,845 | 9 | In-hospital mortality |
| PhysioNet 2012 | 7,993 | 10 | In-hospital mortality |

Table 1: Primary original-cohort CliniCause resources. Counts refer to analysis records and admitted non-chronic proxy exposures.

scribes its three pillars: proxy recoverability, cross-estimator agreement, and clinical-literature corroboration.

### 4.1 Predictive Recoverability

We first ask whether the knowledge-based proxy rules are supported by patterns in the observed ICU time series. The accepted LLM-elicited rules assign each proxy a binary value. This yields nine binary prediction targets in MIMIC-III and ten in PhysioNet 2012, one for each admitted exposure. We train four models designed for irregular time series—STraTS, GRU, GRU-D, and TCN—to predict these rule-derived labels from the underlying patient measurements (Tipirneni and Reddy 2022; Cho et al. 2014; Che et al. 2018; Bai, Kolter, and Koltun 2018).

We report macro AUROC, AUPRC over a 3-fold CV test.

### 4.2 Effect Estimation and Stability Metrics

We evaluate all 19 admitted dataset–exposure comparisons: nine in MIMIC-III and ten in PhysioNet. In each comparison, one binary proxy $Z_k$ is designated as the exposure $T_k$, source-observed in-hospital mortality is the outcome $Y$, and the dataset-specific DAG determines the observed adjustment variables. The datasets are analyzed separately, and all prespecified comparisons are retained, including weak, negative, and estimator-discordant results.

We apply CausalForestDML, LinearDML, and CausalPFN. CausalForestDML combines orthogonalized nuisance estimation with a causal-forest final stage, whereas LinearDML uses a linear final effect model (Chernozhukov et al. 2018; Wager and Athey 2018; Athey, Tibshirani, and Wager 2019; Oprescu et al. 2019). CausalPFN provides a distinct pretrained estimation paradigm (Meresht et al. 2025). The DML estimators receive adjustment and effect-modification variables through separate inputs; CausalPFN receives their deduplicated union.

For estimator $e$, exposure $k$, and dataset $d$, we report the mean model-estimated CATE

$$\widehat{\mu}_{ek}^{(d)} = \frac{1}{n_d} \sum_{i=1}^{n_d} \widehat{\tau}_{ek}^{(d)}(i).$$

We assess stability using sign agreement, pairwise Spearman correlations between within-dataset exposure rankings, and pairwise estimator RMSE,

$$\mathrm{RMSE}_{e,e'}^{(d)} = \sqrt{\frac{1}{K_d} \sum_{k=1}^{K_d} \left( \widehat{\mu}_{ek}^{(d)} - \widehat{\mu}_{e'k}^{(d)} \right)^2}.$$

These quantities measure agreement between estimators, not error relative to causal ground truth.

As disruption checks, we repeat the DML analyses after independently permuting either the exposure or mortality outcome ten times. We compare the observed estimate with the corresponding permutation distribution to test whether the fitted pattern disappears when the empirical exposure–outcome structure is broken. These tests serve as sanity checks for possible data leakage, pipeline artifacts, or other unintended structure that could produce apparent effects after the exposure–outcome relationship has been broken.

Finally, we apply EconML's nonparametric omitted-variable sensitivity analysis to the two DML estimators. For each comparison, we report the confidence-interval robustness value: the largest equal confounding strength $c_T = c_Y = c$, under worst-case alignment ($\rho = 1$), for which the sensitivity interval remains separated from zero. Here, $c$ denotes the fraction of residual variation in both exposure and outcome explained by a hypothetical unobserved confounder. Smaller values therefore indicate greater fragility to omitted confounding. This diagnostic is limited to the DML estimators because it requires EconML's fitted nuisance models and residualized exposure and outcome quantities, which CausalPFN does not expose.

### 4.3 Clinical-Evidence Corroboration

After resource construction and causal estimation, we conducted a separate literature audit to assess whether the estimated effects were broadly compatible with mortality patterns reported for corresponding clinical constructs. For each CliniCause proxy, we searched for ICU studies examining the nearest clinically recognized state and reporting a numerical mortality contrast. For example, renal dysfunction was mapped to acute kidney injury defined using creatinine or urine-output criteria, while shock was mapped to comparisons of septic shock with sepsis without shock. These mappings are construct-level analogies; they do not imply that the published exposure and the CliniCause are identical.

Candidate studies were identified through parallel AI-assisted literature searches using the common query reported in the supplementary material Section 4, and were subsequently checked against the cited publications (specifically, we used NotebookLM, Perplexity, and alphaXiv)

We prioritized explicit causal analyses, followed by matched or adjusted observational comparisons and, when no stronger comparator was available, unadjusted mortality contrasts. Numerical comparisons were made only when absolute mortality differences were reported or directly recoverable; odds ratios and hazard ratios were retained on their original scales. Agreements and discrepancies were recorded symmetrically. Because the studies differ in construct definition, population, adjustment strategy, and outcome horizon, their results are treated as contextual clinical evidence rather than causal ground truth or calibration targets.

## 5 Results

In this section, we report the empirical evaluation described in Section 4; Table 1 summarizes the two CliniCause resources, and the dataset-specific DAGs are provided in the supplementary material. Experiments used one GPU, eight

| Dataset | Model | Test AUROC |
|---|---|---|
| *Task 1: Proxy recovery from irregular time series* | | |
| MIMIC-III | STraTS | **0.905** |
| | GRU | 0.881 |
| | GRU-D | 0.884 |
| | TCN | 0.867 |
| PhysioNet 2012 | STraTS | 0.915 |
| | GRU | 0.915 |
| | GRU-D | **0.918** |
| | TCN | 0.899 |
| *Task 2: Mortality prediction from knowledge-defined proxies* | | |
| MIMIC-III | Logistic regression | 0.826 |
| | MLP | **0.831** |
| PhysioNet 2012 | Logistic regression | 0.776 |
| | MLP | **0.778** |

Table 2: Held-out predictive evaluation of the constructed representation. Task 1 reports macro AUROC for recovering the rule-derived proxy vector from irregular time series. Task 2 reports AUROC for predicting source-observed in-hospital mortality from the majority-vote (MAJ) proxy vector. Bold denotes the highest archived AUROC within each dataset and task. These results assess predictive information, not causal validity.

CPU cores, and 29 GB of memory, with support for a two-GPU, 16-core, 55 GB configuration. The temporal models were adapted from the public STraTS implementation for multi-label proxy prediction. CausalForestDML and LinearDML were run through `econml.dml`, and CausalPFN through its official implementation. The remaining resource construction, evaluation, and matching code was implemented by the authors.

### 5.1 Construct Plausibility and Predictive Recoverability

The informal ICU consultations found the selected constructs and graph relationships broadly plausible, but produced no recorded ratings or independent adjudication statistics.

We next evaluate the constructed representation through two complementary predictive tasks (Table 2). First, irregular time-series models recover the knowledge-defined proxy values with held-out macro AUROC values of 0.867–0.918, indicating that the knowledge-defined rules and cutoffs correspond to patterns recoverable from the underlying ICU measurements. Second, using only the knowledge-defined proxy vector, mortality-prediction AUROC reaches 0.831 in MIMIC-III and 0.778 in PhysioNet. For context, STraTS reports mortality-prediction AUROCs of approximately 0.89 and 0.84, respectively, when operating directly on the full irregular time series (Tipirneni and Reddy 2022). Thus, the compact proxy representation retains substantial mortality-relevant information.

### 5.2 Cross-Estimator Stability and Sensitivity

We next evaluate whether the estimated effect patterns remain stable across estimators and diagnostic perturbations.

| Dataset | Sign agreement | Spearman $\rho$ | Pairwise RMSE |
|---|---|---|---|
| MIMIC-III | 9/9 | 0.983–1.000 | 1.35–2.57 pp |
| PhysioNet 2012 | 9/10 | 0.794–0.964 | 1.08–2.04 pp |

Table 3: Cross-estimator stability. Sign agreement requires all three estimators to agree; correlations and RMSE values give the ranges across estimator pairs. RMSE measures disagreement, not error against causal ground truth.

The main text summarizes agreement in direction, ranking, and magnitude; the complete exposure-level estimates and diagnostics are reported in Section 1 of the supplementary material.

As summarized in Table 3, CausalForestDML and LinearDML agreed in direction in all 19 comparisons, and all three estimators agreed in 18 of 19. The sole disagreement was PhysioNet shock, for which the DML estimates were negative and the CausalPFN estimate was slightly positive.

Within-dataset rankings were very stable in MIMIC-III and moderately to highly stable in PhysioNet. Pairwise RMSE ranged from 1.08 to 2.57 percentage points, although larger exposure-specific differences occurred for MIMIC-III cardiac strain and PhysioNet hepatic dysfunction, shock, and renal dysfunction. Across the nine approximately shared constructs, eight had the same direction, but their average-effect rankings were only moderately correlated ($\rho = 0.533$). Thus, agreement is strongest across estimators applied to a fixed resource, not across resources with different cohorts, proxies, DAGs, and adjustment sets.

*Permutation tests* provided a complementary disruption check. After permuting either the exposure or outcome, the fitted mean effects were close to zero. The observed estimate lay more than two permutation standard deviations from the permuted mean in all 36 MIMIC-III DML comparisons and in 34 of 40 PhysioNet DML comparisons. The exceptions were concentrated in PhysioNet shock and coagulation/hematologic dysfunction.

*The omitted-variable analysis* indicates relatively weak robustness. PhysioNet shock and coagulation/hematologic dysfunction already have confidence intervals containing zero, and among the remaining 17 comparisons, a worst-case-aligned unobserved confounder explaining only about $1\%$ of the residual variation in both exposure and mortality is sufficient to move the first additional interval to zero. Such a small amount of omitted explanatory power is plausible in observational ICU data, where no measured representation can reasonably be expected to capture every determinant of clinical state and mortality. At the stronger $5\%$-by-$5\%$ scenario, only 12 of 19 intervals remain separated from zero. Thus, the DML conclusions are not uniformly robust under this specific sensitivity model. This fragility itself demonstrates the value of CliniCause as an observational testbed: future methods can be evaluated on whether they detect, quantify, or better withstand modest unobserved confounding, rather than only recovering effects in benchmarks whose causal mechanisms are fully specified.

| Comparison pattern | Clinical constructs |
|---|---|
| Similar direction and scale | Renal/AKI; hepatic/bilirubin; cardiac injury |
| Dataset dependent | Inflammation/sepsis: close in PhysioNet, larger in MIMIC-III |
| Substantial scale or directional discrepancy | Shock; coagulation; respiratory; neurologic; metabolic |
| No comparable numerical contrast | Global severity |

Table 4: Qualitative synthesis of the clinical-evidence comparison. Categories summarize approximate direction and scale; the clinical studies do not provide ground-truth CATEs. Full numerical comparisons, study designs, mortality horizons, and citations are reported in the supplement.

### 5.3 Clinical-Evidence Corroboration

We applied the clinical-evidence validation protocol described in Section 4.3. Table 4 summarizes the comparison, with the complete numerical evidence audit reported in the supplementary material, Section 2. The renal comparison provides the closest match: a published attributable 30-day mortality estimate of $8.1$ percentage points is close to the CliniCause ranges of $8.7$–$9.1$ in MIMIC-III and $9.1$–$12.0$ in PhysioNet. Hepatic and cardiac estimates were also similar in direction and approximate scale, while inflammation/sepsis aligned more closely in PhysioNet than in MIMIC-III.

Shock was the clearest discrepancy. The published mortality contrast was $11.1$ points, compared with $2.1$–$3.2$ in MIMIC-III and $-2.7$–$0.4$ in PhysioNet. Coagulation, respiratory, neurologic, and metabolic estimates also differed substantially from their published comparators. These differences may reflect construct severity, proxy thresholds, temporal windows, population and outcome differences, adjustment choices, or estimator behavior. They identify priorities for targeted sensitivity analyses rather than grounds for selecting or discarding results.

## 6 Discussion and Limitations

CliniCause demonstrates a representation-centered alternative to constructing causal benchmarks. Rather than simulate exposure assignments or outcomes, the pipeline preserves source-observed clinical processes and places researcher intervention in an inspectable layer of proxy definitions, temporal rules, causal assumptions, and adjustment choices. This approach is not limited to ICU records. Public longitudinal data in education, transportation, environmental monitoring, energy systems, and industrial telemetry may offer similar opportunities wherever domain knowledge can map raw measurements to meaningful candidate exposures and outcomes.

A natural next step is to turn these resources into an expanding evaluation system for published causal methods. The present analysis already demonstrates this potential: despite strong cross-estimator agreement, omitted-variable sensitivity analysis reveals that several DML conclusions are fragile to modest unobserved confounding. Continually adding new resources would enable methods to be stress-tested across previously unseen populations, representations, support conditions, and assumption violations, reducing adaptation to the small set of established benchmarks. Patient-level test data and outcome summaries could remain hidden, with submitted estimators evaluated through a controlled interface. Such infrastructure would reveal where methods generalize, disagree, or fail without allowing developers to tune directly to every evaluation dataset.

CliniCause complements rather than replaces existing benchmarks. Synthetic and semi-synthetic designs provide exact effects under specified data-generating mechanisms; RCT-derived evaluations provide experimental anchors but depend on sampling and transport assumptions; and data-fusion benchmarks depend on the comparability of their source populations. These resources make accuracy measurable, but performance remains conditional on researcher-specified data-generating, sampling, or data-fusion mechanisms. CliniCause makes the opposite tradeoff: it gives up a known answer key in exchange for evaluation under source-observed measurement, support, missingness, and outcome processes.

Several limitations bound the present contribution. The proxy states are operational analytical constructs rather than chart-adjudicated phenotypes or literal interventions, and physician feedback was informal rather than a documented independent review. The LLM-assisted design was not compared with clinician-only, literature-only, or alternative-LLM construction. MIMIC-III and PhysioNet access conditions restrict redistribution and complicate independent reproduction. Finally, estimator agreement, predictive recoverability, and clinical-literature corroboration provide converging but non-identifying evidence. They do not establish proxy validity, graph correctness, causal identification, or external clinical validity, and the resulting estimates must not be interpreted as treatment recommendations. This limitation is not unique to CliniCause: synthetic and semi-synthetic benchmarks provide causal truth within their specified mechanisms, but whether that constructed truth reflects causal processes operating in nature remains an open question for the field.

## Code and Data Availability

The preliminary source code is included with the submission for review and will be released publicly upon acceptance. Patient-level data are not redistributed. MIMIC-III is available through PhysioNet under credentialed access: users must complete approved human-subjects research training and sign the PhysioNet data use agreement. The PhysioNet/Computing in Cardiology Challenge 2012 data are publicly available under the Open Data Commons Attribution License v1.0. Accordingly, authorized users can obtain both source datasets directly from PhysioNet and apply the released CliniCause pipeline locally.

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